

1. Given the data as listed below

x	4.0	4.2	4.5	4.7	5.1	5.5	5.9	6.3
y	102.6	113.2	130.1	142.1	167.5	195.1	224.9	256.8

- Construct the least squares approximation of degree two and compute the error.
- Construct the least squares approximation of the form be^{ax} and compute the error.
- Construct the least squares approximation of the form bx^n and compute the error.

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op/數值方法/hw8/q1.py
=== (a) Degree 2 Polynomial ===
Model: y = 6.6912x^2 + -1.8837x + 3.0864
Squared Error: 0.0052

=== (b) Exponential Fit y = b * e^(a*x) ===
Model: y = 21.4445 * e^(0.3985x)
Squared Error: 94.9830

=== (c) Power Fit y = b * x^a ===
Model: y = 6.2390 * x^2.0196
Squared Error: 0.0117
```

compute the error.

2. Find the least squares polynomial approximation of degree two on the interval $[-1,1]$ for the function $f(x) = \frac{1}{2}\cos x + \frac{1}{4}\sin 2x$

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op/數值方法/hw8/q2.py
Final result:
P2(x) = 0.498279 + 0.326548x + -0.232631x^2
Least square error: 3.240431e-03
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3. Determine the discrete least squares trigonometric polynomial S_4

using $m=16$ for $f(x) = x^2 \sin x$ on the interval $[0,1]$.

b. Compute $\int_0^1 S_4(x) dx$

c. Compare the integral in part (b) to $\int_0^1 x^2 \sin x dx$

d. Compute the error $E(S_4)$

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op/數值方法/hw8/q3.py
=== S_4(x) approximation results ===
∫01 S4(x) dx ≈ 0.210277
∫01 x2 sin(x) dx ≈ 0.223244
Discrete least squares error E(S4) = 0.218015
PS C:\Users\User\OneDrive\Desktop\數值方法\hw8>
```